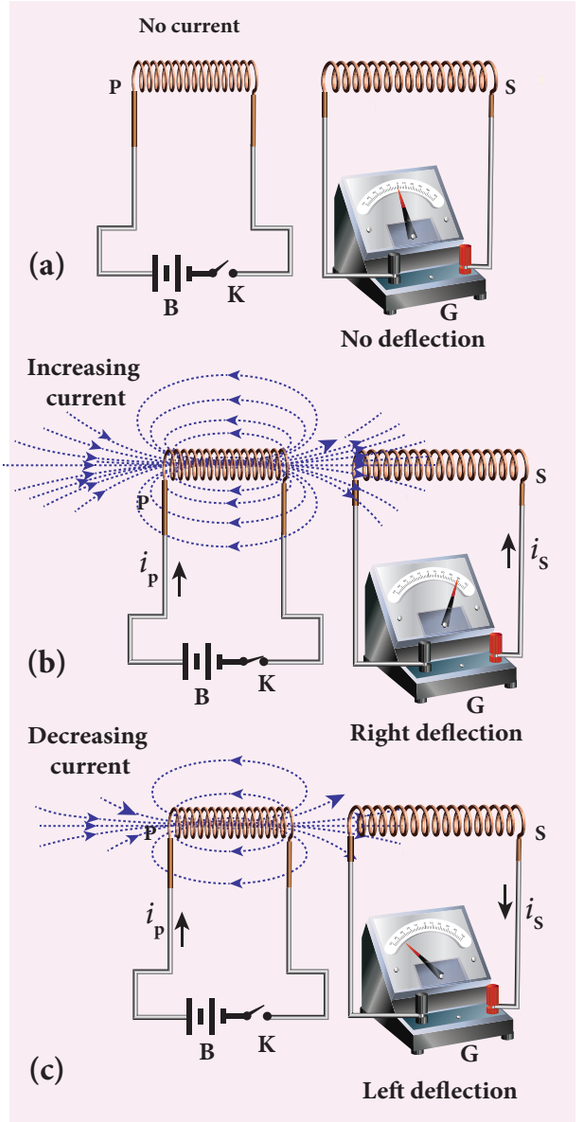


Figure 4.5 Explanation of Faraday's second experiment

Similarly, when the primary circuit is broken, the decreasing primary current induces an electric current in the secondary coil, but in the opposite direction (Figure 4.5(c)). So there is deflection in the galvanometer whenever there is a change in the primary current.

Note

The symbol ε is used for permittivity in unit 1 and for emf in this chapter. Reader is advised to use ε in relevant context.

Importance of Electromagnetic Induction!

The application of the phenomenon of Electromagnetic Induction is almost everywhere in the present day life. Right from home appliances to huge factory machineries, from cellphone to computers and internet, from electric guitar to satellite communication, all need electricity for their operation. There is an ever growing demand for electric power.

All these are met with the help of electric generators and transformers which function on electromagnetic induction. The modern, sophisticated human life would not be possible without the discovery of electromagnetic induction.

The conclusions of Faraday's experiments are stated as two laws.

First law

Whenever magnetic flux linked with a closed circuit changes, an emf is induced in the circuit.

Second law

The magnitude of induced emf in a closed circuit is equal to the time rate of change of magnetic flux linked with the circuit.

If the magnetic flux linked with the coil changes by $d\Phi_B$ in a time dt, then the induced emf is given by

$$\varepsilon = -\frac{d\Phi_B}{dt}$$

The negative sign in the above equation gives the direction of the induced current which will be dealt with in the next section.

If a coil consisting of N turns is tightly wound such that each turn covers the same

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